

CNF

1) Convert the following CFG to CNF (Chomsky Normal form)
[Assume the first rule to contain the start the variable]

a. $W' \rightarrow W$
 $W \rightarrow TWY \mid t \mid TW \mid WY$
 $T \rightarrow tTW \mid t \mid tW$
 $Y \rightarrow WyW \mid T \mid yy$

b. $S' \rightarrow S$
 $S \rightarrow RSR \mid rT \mid RS \mid SR \mid r$
 $R \rightarrow T \mid S$
 $T \rightarrow t$

c. $S \rightarrow A \mid B$
 $A \rightarrow aAs \mid a \mid bb \mid AS$
 $B \rightarrow A \mid aa \mid SbS$
 $C \rightarrow c$

d. $W' \rightarrow W$
 $W \rightarrow TWY \mid t \mid TW \mid WY$
 $T \rightarrow tTW \mid t \mid \epsilon$
 $Y \rightarrow WyW \mid T \mid yy$

e. $S \rightarrow A \mid B$
 $A \rightarrow aAs \mid a \mid bb \mid \epsilon$
 $B \rightarrow A \mid aa \mid SBS \mid \epsilon$
 $C \rightarrow c$

f. $S' \rightarrow S$
 $S \rightarrow RSR \mid rT \mid RS \mid SR \mid \epsilon$
 $R \rightarrow T$
 $T \rightarrow t$

g. $S \rightarrow aAa \mid bBb \mid aa$
 $A \rightarrow a \mid B \rightarrow b$
 $C \rightarrow a \mid b \mid ab$

h. $W \rightarrow TW \mid t \mid TW$
 $W' \rightarrow a$
 $T \rightarrow tTW \mid t \mid \epsilon$
 $Y \rightarrow c$

i. $S \rightarrow A \mid B$
 $E \rightarrow A \mid B$
 $A \rightarrow aAs \mid a \mid bb \mid \epsilon$
 $B \rightarrow A \mid aa \mid SBS \mid \epsilon$
 $C \rightarrow \epsilon$
 $D \rightarrow a \mid b$

Question -4) Convert the following CFG to CNF

a) Start symbol S

$S \rightarrow aAa \mid bBb \mid \epsilon$
 $A \rightarrow C \mid a$
 $B \rightarrow C \mid b$
 $C \rightarrow CDE \mid \epsilon$
 $D \rightarrow A \mid B \mid ab$

b) Start symbol S

$S \rightarrow AB \mid CA$
 $A \rightarrow a$
 $B \rightarrow BC \mid AB$
 $C \rightarrow aB \mid b$

c) Start symbol S

$S \rightarrow ASB \mid \epsilon$
 $A \rightarrow aAS \mid a$
 $B \rightarrow SbS \mid A \mid bb$

d) Start symbol S

$S \rightarrow 0A0 \mid 1B1 \mid BB$
 $A \rightarrow C$
 $B \rightarrow S \mid A$
 $C \rightarrow S \mid \epsilon$

e) Start symbol S

$S \rightarrow AAA \mid B$
 $A \rightarrow aA \mid B$
 $B \rightarrow \epsilon$

f) Start symbol S

$$\begin{array}{l} S \rightarrow AB \mid CD \\ A \rightarrow BC \mid a \\ B \rightarrow AC \mid C \\ C \rightarrow AB \mid CD \\ D \rightarrow AC \mid d \end{array}$$

g) $S \rightarrow aSb \mid b$

CYK

a.

Grammar	Strings to check
$\begin{array}{l} S \rightarrow AB \mid BC \\ A \rightarrow BA \mid a \\ B \rightarrow CC \mid b \\ C \rightarrow AB \mid a \end{array}$	$\begin{array}{l} \text{a) } w = ababb \\ \text{b) } w = baaba \\ \text{c) } w = ababa \\ \text{d) } w = baaab \\ \text{e) } w = aabab \\ \text{f) } w = ababaa \end{array}$

b.

$S_0 \rightarrow WB \mid b$

$S \rightarrow WB \mid b$

$A \rightarrow a$

$B \rightarrow b$

$W \rightarrow AS$

String to check: aabbbb